11. Sequences and Series

11.10. Taylor and Maclaurin Series

11.10 Learning Objectives
- Find the Taylor Series Expansion of a function
- Find the Maclaurin Series Expansion of a function
- Find the function represented by the Taylor or Maclaurin Series
- Use known Maclaurin Series Expansion of functions to describe complicated functions

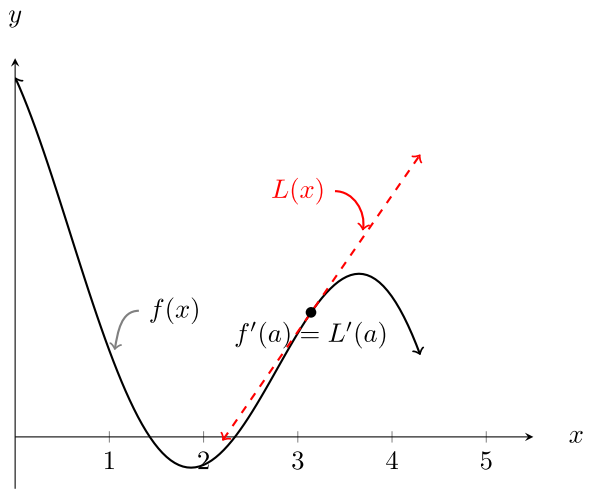
11.10.1. Motivation: Linearization of a function

Recall the linearization of a function

Definition 11.10.1 Linearization of a function

For a (continuous near a) function f, the linearization of f centered at a is the tangent line to the function at the point $(a, f(a))$. Precisely, in *point slope* form $(y = m(x - x_1) + y_1)$

$$L(x) = f'(a)(x - a) + f(a)$$



- The idea behind linearization is to come up with a simple function (a line L) that matches f in function value and first derivative.
- What's better than matching the function in function value and first derivative at $x = a$?
  - Matching the function in value, first derivative, *and* second derivative at $x = a$.
- What's better than matching the function in value, first derivative, and second derivative at $x = a$?
  - Matching the function in value, first derivative, second derivative, *and* third derivative at $x = a$
  - ...
- Then we build up from there! Going one more degree at a time, matching one extra derivative at $x = a$ each term!

11.10.2. Taylor Series and Maclaurin Series

We have been able to write the power series of some functions using algebraic manipulations to line it up with $\frac{1}{1-u}$ type things. However, we may want to represent other functions with power series.

Suppose that f can be represented with a power series. Then,

$$f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + c_4(x-a)^4 + \ldots$$

We already know that when we evaluate this at $x = a$ we get

$$f(a) = c_0$$

Now, taking the element-wise derivative we see that

$$\begin{aligned} f'(x) &= c_1 + 2 \cdot c_2(x-a) + 3 \cdot c_3(x-a)^2 + 4 \cdot c_4(x-a)^3 + \ldots \\ \text{so} \quad f'(a) &= c_1 \end{aligned}$$

Taking another derivative we see

$$\begin{aligned} f''(x) &= 2 \cdot 1 \cdot c_2 + 3 \cdot 2 \cdot c_3(x-a) + 4 \cdot 3 \cdot c_4(x-a)^2 + \ldots \\ \text{so} \quad f'(a) &= 2c_1 \\ \text{or} \quad c_1 &= \frac{f'(a)}{2} \end{aligned}$$

Taking another derivative we see that

$$\begin{aligned} f'''(x) &= 3 \cdot 2 \cdot 1 \cdot c_3 + 4 \cdot 3 \cdot 2 \cdot c_4(x-a) + \ldots \\ \text{so} \quad f'''(a) &= 3!c_3 \\ \text{or} \quad c_3 &= \frac{f'''(a)}{3!} \end{aligned}$$

Doing one more to really see the pattern,

$$\begin{aligned} f^{(4)}(x) &= 4 \cdot 3 \cdot 2 \cdot 1 \cdot c_4 + 5 \cdot 4 \cdot 3 \cdot 2 \cdot c_5(x-a) + \ldots \\ \text{so} \quad f^{(4)}(a) &= 4!c_4 \\ \text{or} \quad c_4 &= \frac{f^{(4)}(a)}{4!} \end{aligned}$$

In general, the nth coefficient must be

$$c_n = \frac{f^{(n)}(a)}{n!}$$

> **Note**
>
> Note that we define $0! = 1$ to make this true when $n = 0$ as well.
>
> Side note: You can also justify this using another application of factorials: In combinatorics, the number of ways to arrange n items in a list is $n!$. If you think about having 0 items to arrange in a list, there is only 1 way to do this, by not putting anything. So $0 \neq 1$ makes sense "physically".

Theorem 11.10.1

If a function f has a power series representation, then

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Note

- Note, this theorem does not say when a function has a power series representation.
- Notation: Recall that $f^{(n)}(x)$ is the nth derivative of $f(x)$, and the parentheses are an important part of this notation. Use this for discussing more than the third or fourth derivative ($f'(x), f''(x), f'''(x)$ are reasonable...). Without the parentheses, $f^n(x)$ is an exponent, meaning $f^n(x) = (f(x)) \cdot (f(x)) \cdot (f(x)) \cdot (f(x)) ... \cdot (f(x)) = (f(x))^n$

Definition 11.10.2 Taylor Series

The series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

is called the **Taylor Series of f centered at a**.

This series is named after Brook Taylor, who wrote a paper using Taylor Series in 1715, but the importance of it wasn't noticed until 1772 when Lagrange stumbled across it.

Definition 11.10.3

If $a = 0$, then we say the series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}x^n$$

is the **Maclaurin Series of f**.

This series is named after Colin Maclaurin, who did a lot of important mathematics using a specialized form of Taylor's series (at $a = 0$). I had fun reading his Wikipedia page, and learning that we have already seen another result of Maclaurin's! And that is Simpson's Rule! (behind the scenes Simpson's Rule comes from integrating a truncated Maclaurin series to approximate the integral!) Also, Maclaurin was a child prodigy who held the record for youngest math professor for centuries until Alia Sabur became a university professor when she was only 18 years old!

Example 11.10.1

Find the Maclaurin series, and its interval of convergence, for the function $f(x) = e^x$.

Solution

We start these problems by finding the first few derivatives evaluated at the center, and try to write a pattern. Note $a = 0$ for Maclaurin series, and we have

$$\begin{aligned} f(x) &= e^x, & f(0) &= e^0 = 1 \\ f'(x) &= e^x, & f'(0) &= e^0 = 1 \\ f''(x) &= e^x, & f''(0) &= e^0 = 1 \\ &\vdots & f^{(n)}(0) &= 1 \quad \text{for all } n \end{aligned}$$

Then we plug the derivatives into

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} x^n \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} x^n \\ e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \end{aligned}$$

Now, we find the interval of convergence. By ratio test,

$$\begin{aligned} L &= \lim_{n\to\infty} \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| \\ &= \lim_{n\to\infty} \left| \frac{x^n x}{(n+1)n!} \cdot \frac{n!}{x^n} \right| \\ &= \lim_{n\to\infty} \frac{|x|}{n+1} \\ &= 0 \end{aligned}$$

So the series converges for all x values, so the radius of convergence is $R = \infty$ and the interval of convergence is $(-\infty, \infty)$.

Example 11.10.2

Find the Taylor series, and its interval of convergence, for the function $f(x) = b^x$ centered at $x = 1$, where $b > 1$.

Solution

First we write out as many derivatives as it takes to identify a pattern

$$\begin{aligned} f(x) &= b^x, & f(1) &= b^1 = b(\ln(b))^0 \\ f'(x) &= \ln(b) b^x, & f'(1) &= \ln(b) b^1 = b(\ln(b))^1 \\ f''(x) &= (\ln(b))^2 b^x, & f''(1) &= (\ln(b))^2 b^1 = b(\ln(b))^2 \\ &\vdots \\ f^{(n)}(x) &= (\ln(b))^n b^x, & f^{(n)}(1) &= b(\ln(b))^n \quad \text{for all } n \end{aligned}$$

Then, we plug this into

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} x^n \\ b^x &= \sum_{n=0}^{\infty} \frac{b(\ln(b))^n}{n!} x^n \end{aligned}$$

Then, to discuss convergence we check ratio test again,

$$
\begin{aligned}
L &= \lim_{n\to\infty}\left|\frac{b(\ln(b))^{n+1}x^{n+1}}{(n+1)!}\cdot\frac{n!}{b(\ln(b))^n x^n}\right| \\
&= \lim_{n\to\infty}\left|\frac{b(\ln(b))^{n}\ln(b)x^{n}x}{(n+1)n!}\cdot\frac{n!}{b(\ln(b))^n x^n}\right| \\
&= \lim_{n\to\infty}\left|\frac{\ln(b)x}{(n+1)}\right| \\
&= \lim_{n\to\infty}\frac{\ln(b)}{n+1}|x| = 0 < 1
\end{aligned}
$$

Again, no matter the x value, the limit is 0, so ratio test says the series converges. Then, the series has radius of convergence $R = \infty$ and interval of convergence $(-\infty, \infty)$

Example 11.10.3

Find the Taylor Series Expansion for $f(x) = \cos(x)$ centered at $x = \pi$, and find its radius of convergence.

Solution

Again we start with derivatives until we find a pattern

$$
\begin{aligned}
f(x) &= \cos(x), & f(\pi) &= \cos(\pi) = -1 \\
f'(x) &= -\sin(x), & f'(\pi) &= -\sin(\pi) = 0 \\
f''(x) &= -\cos(x), & f''(\pi) &= -\cos(\pi) = 1 \\
f'''(x) &= \sin(x), & f'''(\pi) &= \sin(\pi) = 0 \\
f^{(4)}(x) &= \cos(x), & f^{(4)}(\pi) &= \cos(\pi) = -1 \leftarrow \text{same as the first value, and will repeat} \\
&\vdots
\end{aligned}
$$

Writing the pattern here is a little harder, let's write a few terms in the series

$$
f(x) = \frac{-1(x-\pi)^0}{1} + 0 + \frac{1(x-\pi)^2}{2!} + 0 + \frac{-1(x-a)^4}{4!} + 0 + \ldots
$$

We only have even terms in the exponent and factorial in the denominator, and even numbers can be represented by $2n$. So we have the Taylor Series centered at $x = \pi$ is

$$
\cos(x) = \sum_{n=0}^{\infty}\frac{(-1)^{n+1}(x-\pi)^{2n}}{(2n)!}
$$

By ratio test we can find the radius of convergence

$$
\begin{aligned}
L &= \lim_{n\to\infty}\left|\frac{(x-\pi)^{2(n+1)}}{(2(n+1))!}\cdot\frac{(2n)!}{(x-\pi)^{2n}}\right| \\
&= \lim_{n\to\infty}\left|\frac{(x-\pi)^{2n+2}}{(2n+2)!}\cdot\frac{(2n)!}{(x-\pi)^{2n}}\right| \\
&= \lim_{n\to\infty}\left|\frac{(x-\pi)^2}{(2n+2)(2n+1)}\right| = 0 < 1
\end{aligned}
$$

So again we have $R = \infty$ and interval of convergence is $(-\infty, \infty)$.

11.10a Section Summary:

- We found the Taylor Series and Maclaurin Series expansions of functions.
- We used ratio test to find the radius and interval of convergence of these series.